WEEK 3 REPORT

Endpoint Monitoring, Log Analysis & network Intrusion Detection

Cybersecurity Internship Weekly Report

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Reporting Period	Week 3
Focus	Endpoint Monitoring, Log Analysis & network Intrusion Detection
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1. Introduction

During Week 3 of the cybersecurity internship, I focused on understanding endpoint monitoring, Linux auditing, network intrusion detection and security log analysis.

The main objective was to understand how SOC analysts monitor Windows and Linux systems, identify suspicious activities and detect potential network attacks.

I worked with different security tools, including Sysmon, Linux auditd and Snort. I also explored security monitoring concepts, SIEM platforms and MITRE ATT&CK techniques.

This week's activities helped me understand how endpoint and network logs are collected, monitored and analyzed during security investigations.

2. Objectives

The main objectives of Week 3 were:

- Understand Endpoint Detection and Response (EDR).
- Learn Windows endpoint monitoring using Sysmon.
- Understand Linux system auditing using auditd.
- Identify and analyze important Linux log files.
- Install and configure Snort IDS/IPS.
- Create custom Snort detection rules.
- Detect network scanning activities using Nmap.
- Understand MITRE ATT&CK T1046.
- Learn log ingestion and SIEM monitoring.
- Understand signature-based and anomaly-based detection.
- Develop practical SOC monitoring and investigation skills.

3. Theoretical Learning

I completed the theoretical learning activities covering the following topics.

3.1 Endpoint Detection and Response (EDR)

- Learned what EDR is and why it is important in cybersecurity.
- Understood how endpoint data helps SOC analysts investigate suspicious activities.
- Studied the difference between traditional antivirus and EDR.
- Learned how endpoint monitoring helps identify malicious processes, file changes and network connections.

3.2 Windows Sysmon

- Learned about Microsoft Sysmon and its functionality.
- Understood how Sysmon records detailed Windows system activities.
- Studied important Sysmon Event IDs.

Important Event IDs:

Event ID	Description
1	Process Create
3	Network Connection
11	File Create
13	Registry Value Set
22	DNS Query

3.3 Linux auditd

- Learned the purpose of Linux auditd.
- Understood how auditd monitors system activities.
- Studied audit logs and their important fields.
- Learned how to search audit records using ausearch.
- Understood the difference between auditd and syslog.

3.4 Snort IDS/IPS

- Learned the working of Snort.
- Studied IDS and IPS modes.
- Understood how Snort analyzes network packets.
- Learned how detection rules generate alerts.
- Studied Snort rule structure and important rule options.

3.5 MITRE ATT&CK T1046

- Studied Network Service Discovery.
- Understood how attackers identify available network services.
- Learned how Nmap scanning activity can be detected.
- Understood how Snort detection rules can be mapped to MITRE ATT&CK techniques.

3.6 Other Topics

I also studied:

- Signature-based detection.
- Anomaly-based detection.
- Windows Event Logs vs Sysmon vs Linux auditd.
- Important Linux log files.
- Log ingestion.
- SIEM monitoring using Wazuh and Splunk.

4. TryHackMe Practical Learning

I completed the following TryHackMe rooms to improve my understanding of Linux logging, log analysis and network intrusion detection

No.	Room Name	Learning Outcome
1	Linux Logging for SOC	Learned Linux logging concepts and important log files.
2	Intro to Logs	Understood log generation and log structures.
3	Intro to Log Analysis	Learned how to examine and interpret security logs.
4	Snort	Learned Snort fundamentals and network intrusion detection.

Separate writeups were prepared for the completed rooms.

5. LetsDefend Practical Learning

I completed the following LetsDefend training lessons.

No.	Lesson Name	Learning Outcome
1	How to Investigate a SIEM Alert	Learned the basic SIEM alert investigation process.
2	Security Solutions	Learned about different cybersecurity security solutions.
3	Network Log Analysis	Understood network log analysis.
4	SIEM 101	Learned SIEM fundamentals and monitoring concepts.
5	Incident Management	Understood security incident management.
6	Incident Response on Linux	Learned Linux incident response fundamentals.

6. Hands-On Practical Implementation

6.1 Installing Sysmon on Windows

Objective: Install Sysmon to monitor Windows endpoint activities.

Activities performed:

- Downloaded Sysmon from the Microsoft Sysinternals source.
- Extracted the installation files.
- Downloaded the SwiftOnSecurity Sysmon configuration.
- Installed Sysmon using an elevated Command Prompt.
- Verified Sysmon installation through Windows Event Viewer.
- Accessed the Sysmon Operational event log.

Learning Outcome:

Understood how Sysmon collects detailed endpoint information and records system activities for security monitoring.

6.2 Generating and Analyzing Sysmon Events

Performed different activities on the Windows VM to generate Sysmon events.

Activities:

- Opened PowerShell.
- Created a new user account.
- Checked network connections.
- Created a text file.
- Located the corresponding events in Event Viewer.

Important events analyzed:

- Event ID 1 – Process Create.
- Event ID 3 – Network Connection.
- Event ID 11 – File Create.

For each event, the relevant details included:

- Event ID.
- Timestamp.
- Process image.
- Command line.
- User information.

- Source and destination IP addresses, where available.
- Target filename, where applicable.

Learning Outcome:

Learned how SOC analysts use Sysmon events to investigate process execution, network connections and file creation activities.

6.3 Linux Auditing Using auditd

Objective: Monitor and analyze Linux system activities using auditd.

Activities performed:

- Installed auditd on the Ubuntu VM.
- Enabled and started the auditd service.
- Verified the service status.
- Executed Linux commands.
- Used ausearch to search audit records.
- Examined audit log entries.

Learning Outcome:

Learned how Linux audit logs record system activities and how security analysts investigate command execution and user activity

6.4 Installing and Configuring Snort

Objective: Configure Snort to monitor network traffic and detect suspicious activities.

Activities performed:

- Updated Ubuntu package repositories.
- Installed Snort and its required packages.
- Verified Snort installation.
- Identified the network interface using ip a.
- Configured HOME_NET and EXTERNAL_NET.
- Opened the local.rules configuration file.
- Added custom detection rules.
- Tested the Snort configuration.

Custom Snort Rules

Rule 1 – ICMP Ping Detection

alert icmp any any -> $HOME_NET any (msg:"ICMP Ping Detected"; sid:1000001; rev:1;)

Rule 2 – SSH Connection Attempt

alert tcp any any -> $HOME_NET 22 (msg:"SSH Connection Attempt"; sid:1000002; rev:1;)

Rule 3 – HTTP Traffic Detection

alert tcp any any -> $HOME_NET 80 (msg:"HTTP Traffic"; sid:1000003; rev:1;)

Learning Outcome:

Learned how Snort inspects network traffic and generates alerts based on configured detection rules.

6.5 Nmap SYN Scan Detection

Objective: Detect network scanning activity using Snort and map the detection to MITRE ATT&CK T1046.

Created an advanced Snort rule to identify repeated TCP SYN packets.

Important rule components:

- flags:S – Matches TCP SYN packets.
- threshold – Controls alert frequency.
- classtype:attempted-recon – Classifies reconnaissance activity.
- metadata:mitre_technique T1046 – Associates the rule with MITRE ATT&CK T1046.
- sid:1000004 – Identifies the custom rule.

Test activity:

- Used Nmap from Kali Linux.
- Performed a SYN scan against the lab target.
- Monitored Snort alerts.
- Examined the detection rule and its MITRE mapping.

Learning Outcome:

Understood how network scanning activities can be detected through Snort rules and how security detections can be mapped to MITRE ATT&CK.

7. Security Event ID and Rule Cheat Sheet

Prepared a cheat sheet containing more than 15 important Windows, Sysmon, Linux auditd and Snort events.

The cheat sheet includes:

- Event ID or Rule.
- Source.
- What it means.
- When to use it.

Important examples:

Event ID / Rule	Source	Description
4624	Windows Security	Successful logon
4625	Windows Security	Failed logon
4688	Windows Security	Process creation
4720	Windows Security	New user account
1	Sysmon	Process Create
3	Sysmon	Network Connection
11	Sysmon	File Create
22	Sysmon	DNS Query
EXECVE	auditd	Command execution
USER_LOGIN	auditd	User login
USER_AUTH	auditd	Authentication attempt
1000001	Snort	ICMP Ping Detection
1000002	Snort	SSH Connection Attempt
1000003	Snort	HTTP Traffic
1000004	Snort	Nmap SYN Scan Detection

8. Tools and Technologies Used

Tool / Technology	Purpose
Windows 10/11	Endpoint monitoring environment
Sysmon	Windows event monitoring
Event Viewer	Windows log analysis
Ubuntu Linux	Linux auditing environment
auditd	Linux security auditing
ausearch	Searching audit records
Snort	Network intrusion detection
Kali Linux	Security testing and Nmap scanning
Nmap	Network discovery and port scanning
TryHackMe	Cybersecurity practical learning
LetsDefend	SOC analyst training
Wazuh / Splunk	SIEM concepts and monitoring

9. Skills Developed

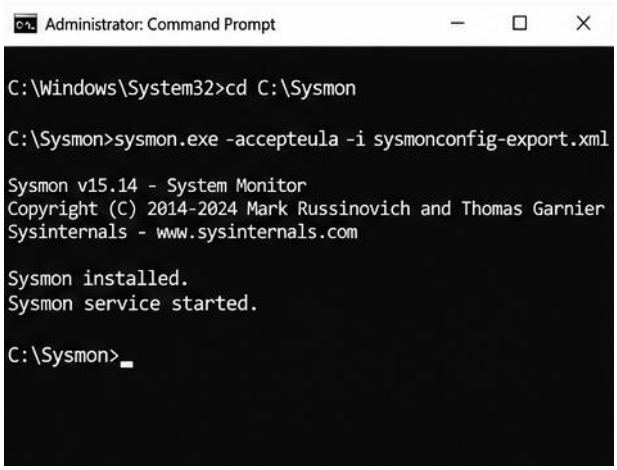
During Week 3, I improved my knowledge and practical understanding of:

- Windows endpoint monitoring.
- Linux log analysis.
- Sysmon event investigation.
- Linux audit log analysis.
- Network intrusion detection.
- Snort rule creation.
- Nmap scanning detection.
- MITRE ATT&CK mapping.
- SIEM fundamentals.
- SOC monitoring and incident investigation.

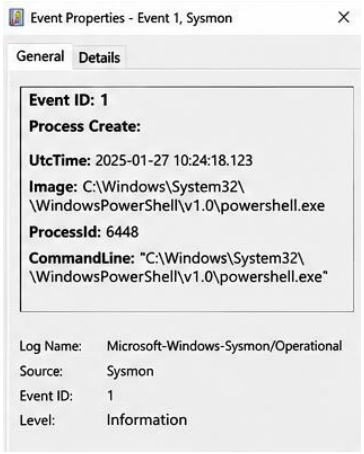
10. Practical Findings and Evidence

The following evidence should be included in the final submission:

1. **Sysmon installation screenshot.**



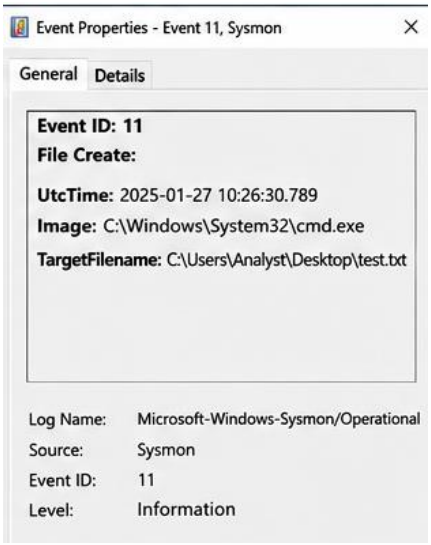
2. **Sysmon Event ID 1 screenshot.**



3. **Sysmon Event ID 3 screenshot.**



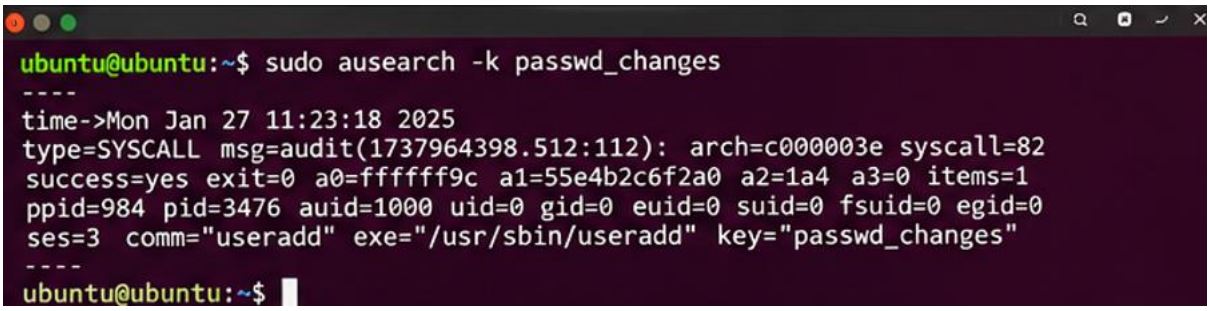
4. **Sysmon Event ID 11 screenshot.**



5. **Linux auditd installation and service status screenshot.**



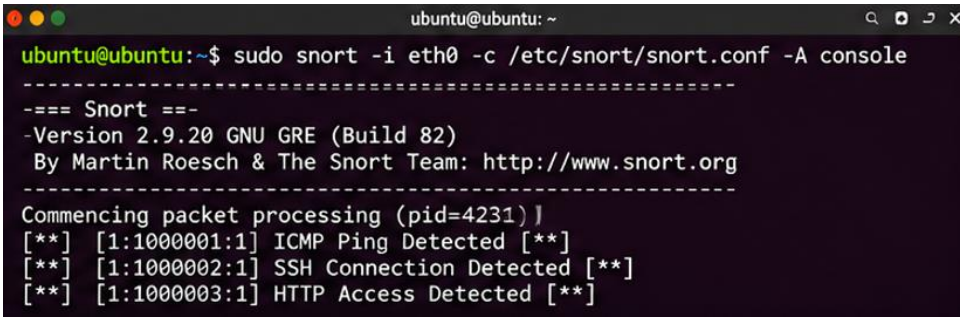
6. **auditd ausearch output screenshot.**



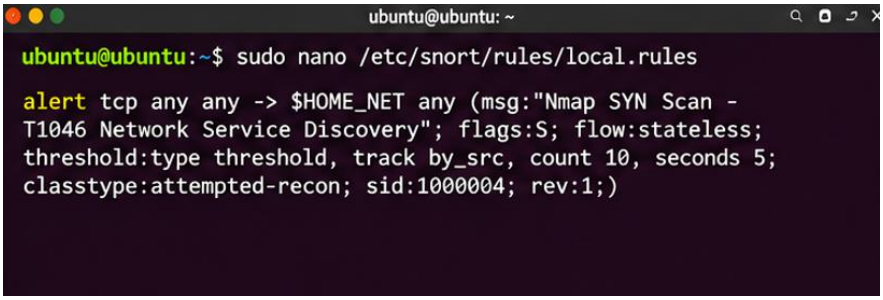
7. **Snort installation and configuration screenshot.**



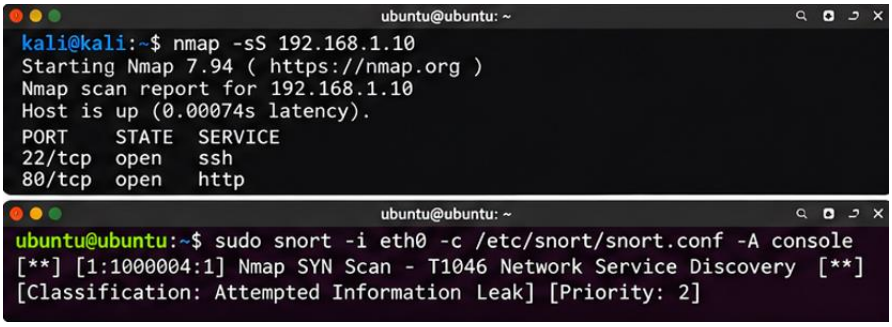
8. **Snort ICMP , SSH , HTTP alert screenshot.**



9. **Nmap SYN scan output**



10. **Snort T1046 detection alert.**



11. Conclusion

Week 3 provided practical exposure to endpoint monitoring, Linux auditing and network intrusion detection.

Through Sysmon, I learned how to monitor Windows endpoint activities and analyze detailed system events. Through auditd, I learned how Linux records security-related activities and how audit records can be searched and investigated.

I also worked with Snort to understand network intrusion detection, create custom detection rules and identify Nmap SYN scanning activities.

Additionally, TryHackMe and LetsDefend helped strengthen my understanding of Linux logging, SIEM alert investigation, network log analysis and incident response.

Overall, this week improved my understanding of SOC operations, endpoint visibility, log analysis and basic threat detection techniques.

12. References

- Microsoft Sysinternals – https://learn.microsoft.com/sysinternals/downloads/sysmon
- SwiftOnSecurity Sysmon Configuration – https://github.com/SwiftOnSecurity/sysmon-config
- Snort Documentation – https://docs.snort.org/
- MITRE ATT&CK T1046 – https://attack.mitre.org/techniques/T1046/
- TryHackMe – https://tryhackme.com/
- LetsDefend – https://letsdefend.io/